1 Explanation certificate: churn100-s2-c2

This section is the explanation certificate of the churn100-s2-c2 network. It is rendered from the canonical JSON record `churn100-s2-c2.json` (schema 1.4.0, SHA-256 `3994ace43ba74682...`), which also holds every evidence payload, the full graphs with weights, and the explanation record. Table 1 summarises the network and its coverage.

Dataset	churn_onehot
Experiment	churn100-s2, seed 2
Genome	`e5394267-5807-44ae-8629-5e157bcfb43a` (NEAT key 24564, generation 99)
Explanation	`4792056e-ee3e-4fde-a1bb-4c4d809e2d61` (98 operations)
Train AUC / accuracy	0.8944 / 0.896 (n = 4000)
Test AUC / accuracy	0.8883 / 0.9 (n = 1000)
Output activation	sigmoid
Phenotype	97 nodes (71 inputs, 25 hidden, 1 output), 89 enabled edges
Explained model state	114 nodes, 91 enabled edges
Refactoring operations	add_identity_node 2, consolidate_node 3, split_node 13
Function preservation	max $\lvert\Delta\,\text{output}\rvert = 5.55 \times 10^{-16}$ on the training split
Annotations	9 (7 leaf, 2 composition)
Structural coverage	1 (113 of 113 candidate nodes; 30 Null)
Compositional coverage	1
Uncovered nodes / edges	0 / 0
Missing compositions	0
Full coverage	yes
Drafted by	AI-assisted draft
Verified by	unverified
Record hash (SHA-256)	`3994ace43ba74682...`

Table 1: Explanation certificate summary: churn100-s2-c2.

Drafter's summary. logit P(churn) = -1.15 + a sparse linear backbone over 33 inputs (day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, ...) + a service-call x low-day-usage gate + a 4+-calls hinge + exactly linear voicemail, state-offset and area-code units, all additive in the logit. The gate is a low-day-minutes hinge whose kink moves up with each customer-service call (median 147 day minutes at 0 calls, 215 at 4, 270 at 6), lowered by high evening charge and switched off in state:0, so with the direct day-minutes edge the logit is V-shaped in day minutes and frequent callers with low usage churn. Service calls act only through hinges. Train AUC 0.894, accuracy 0.896.

What resisted explanation. Nothing structural: no hidden node needed splitting (13 input splits, 3 consolidations). Node 14653 (area code 408) is never active on train (dead); 252 (state:44 / very low international charge) is active on 100 rows with at most +0.03 logit. The gate under-predicts churn for frequent callers with low day usage (CSC>=4 and day minutes <160: 113 rows, model 0.52 vs empirical 0.88). The voicemail chain is exactly linear, and mean-ablating it raises train AUC (0.8944 -> 0.8979): its extra voicemail-plan penalty slightly hurts ranking. closed_form_max_abs_err 0.043 (state_offsets) is 3-significant-figure rounding on state one-hots (standardised values up to 8); the numeric function is exact (2.7e-15).

1.1 Plain-language summary

AI-assisted draft for review (unverified). Each sentence cites the claims (C01, C02, ...) that back it; every cited claim was computed on the model and the training split and passed its automatic checks. Magnitudes are changes in predicted risk for a typical patient; confidence words follow the test-split check and n. The text describes what the model does, not the world.

Summary (drafted). For a typical customer (in West Virginia, area code 415, no international or voice-mail plan, 1 call to customer service and median usage) the model predicts a chance of leaving of about 1%; it adds up six parts: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] This part adds up most inputs with fixed weights, and daytime minutes count most: they raise a typical customer's chance of leaving steadily, from under 1% at 112 to about 11% at 249 minutes. [C053] An international plan takes the chance from about 1% to about 14%, a voice-mail plan lowers it to under 1%, and more voice-mail messages raise it steadily, from about 1% at 0 to about 4% at 32 messages. [C052, C061, C056] A higher international charge, more evening minutes, a higher evening charge and more night-time minutes also raise it steadily, each by one to two points. [C057, C059, C060, C065] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] This part adds up two parts: customer-service calls combined with low daytime use, and a rise from 4 calls upwards. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] This part reads the voice-mail plan and one state: through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] This part reads only the state: it raises a typical customer's chance from about 1% to about 2% in Washington and in Montana, and by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] This part reads the area code and barely changes the predictions (by at most half a point). [C137] This small part reads the international charge and being in Utah, and barely changes the predictions (by at most 0.2 points). [C144]

The whole model. The model adds up six parts into one score: fixed-weight inputs, service calls, a voice-mail discount, state units, an area-code unit and a small Utah unit. [C051] *Evidence.* It is made up of these six parts (each explained below); the score is their sum, which was checked on all 4000 training customers. [C051] Changing one input at a time across the whole model, an international plan takes a typical customer's chance of leaving from about 1% to about 14%, and a voice-mail plan lowers it to under 1%. [C001, C010] In Washington the chance is about 9%, against about 1% in West Virginia. [C003] More customer-service calls raise it slowly at first, then faster, from about 1% at 0 calls to about 2% at 3 calls. [C011] The model ignores the daytime charge, evening calls, international minutes, night-time calls and the night-time charge. [C046, C047, C048, C049, C050]

Fixed-weight inputs. This part adds up most inputs with fixed weights: daytime minutes, an international plan and voice-mail messages raise the chance of leaving, steadily for the minutes and messages, and a voice-mail plan lowers it. [C053, C052, C056, C061] *Evidence.* Through it, daytime minutes take a typical customer steadily from under 1% at 112 to about 11% at 249 minutes, and an international plan from about 1% to about 14%; a voice-mail plan lowers the chance to under 1%. [C053, C052, C061] Through it, the state moves the chance from about 1% in West Virginia to about 5% in Washington and in California, and to under 1% in Hawaii (41 to 122 training customers per state). [C054, C055, C062] Its formula reproduces the part exactly on all 4000 training customers. [C085]

Service calls. This part raises the chance of leaving with calls to customer service, from 4 calls upwards, slowly at first, then faster, and more so for customers with low daytime use, in an uneven way; it adds up two parts. [C103, C099, C098] *Evidence.* It is made of two parts that are added: calls with low usage, and many service calls. [C103] Through it, daytime minutes lower the chance only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), in an uneven way: for a typical such customer, from about 28% at 91 minutes to about 1% at 352 minutes. [C098] Customer-service calls add to it from 4 calls upwards, slowly at first, then faster: for a typical such customer, from about 3% at 0 calls to about 8% at 4 calls. [C099] With an evening charge above 17.12 US dollars, the chance is lowest at about 148 minutes of daytime use and higher either side: from about 1% at 112 minutes to about 13% at 249 minutes, against about 2% to about 9% at or below that charge. [C095]

Calls with low usage. This part raises the chance of leaving for customers with low usage, more so when they call customer service: daytime minutes lower it only while they stay below about 166 minutes (between 145 and 213 minutes for different customers), and calls raise it, both in an uneven way. [C112, C106] *Evidence.* It acts on 3063 of 4000 training customers, 393 of whom left; for the typical customer it acts on, it changes the chance by less than a point. [C117] Through it, a typical customer's chance falls in an uneven way from about 28% at 91 minutes of daytime use to about 1% at 352 minutes, with no further fall past about 166 minutes (between 145 and 213 minutes for different customers). [C112] A higher international charge lowers it steadily up to about 2.95 US dollars (between 2.95 and 4.10 US dollars for different customers), then no further: for a typical such customer, from about 4% to about 1%. [C115] Customer-service calls raise it in an uneven way, from about 1% at 0 calls to about 2% at 3 calls. [C106] Its formula reproduces the part exactly on all 4000 training customers. [C118]

Many service calls. This part raises the chance of leaving from 4 customer-service calls upwards, slowly at first, then faster. [C122] *Evidence.* It acts on 318 of 4000 training customers, 159 of whom left; for the typical customer it acts on, the chance is about 7% without it and about 12% with it. [C123] Through it, the chance rises slowly at first, then faster: going from 0 to 4 calls moves a typical such customer's chance from about 1% to about 2%. [C122] Its formula reproduces the part exactly on all 4000 training customers. [C124]

Voice-mail discount. This part gives customers with a voice-mail plan a further discount on the chance of leaving. [C119] *Evidence.* Through it, a voice-mail plan lowers a typical customer's chance from about 1% to under 1%, and customers in North Carolina get under 1% instead of about 1%. [C119, C120] Its formula reproduces the part exactly on all 4000 training customers. [C121]

State units. This part reads only the state and raises the chance of leaving in a few states, most in Washington and Montana. [C125, C126] *Evidence.* Through it, a typical customer's chance goes from about 1% to about 2% in Washington and in Montana, and up by less than a point in Kentucky and Ohio. [C125, C126, C127, C128] In Arizona, Delaware, North Dakota and Nevada it lowers the chance by less than a tenth of a point. [C129, C130, C131, C132] Its formula reproduces the part exactly on all 4000 training customers. [C133]

Area-code unit. This part reads the area code and barely changes the predictions. [C137] *Evidence.* Through it, a typical customer's chance stays at about 1% in area codes 408 and 510, as in area code 415. [C134, C135] Switching it off changes the predictions by at most half a point. [C137] Its formula reproduces the part exactly on all 4000 training customers. [C138]

Small Utah unit. This small part reads the international charge and being in Utah, and barely changes the predictions. [C144] *Evidence.* Through it, the international charge barely changes the chance, which stays at about 1% for the typical customer it acts on. [C139] Switching it off changes

the predictions by at most 0.2 points. [C144] Its formula reproduces the part exactly on all 4000 training customers. [C145]

Worth checking against what is known. Worth checking: the state moves the chance by several points (for example Washington about 9% and California about 5%, Hawaii under 1%), based on 41 to 122 training customers per state. [C003, C004, C012] Worth checking: the model treats a voice-mail plan as lowering the chance of leaving but more voice-mail messages as raising it. [C010, C005] Worth checking: daytime minutes, the evening charge and the international charge raise the chance through the fixed-weight inputs but lower it through calls with low usage. [C053, C104, C060, C107, C057, C105]

Claims.

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C001	direction: international calling plan: no international plan to an international plan (whole model) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5; range across patients 2.60 to 2.60	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C002	direction: total daytime minutes: 112 minutes to 249 minutes (whole model) *shape: dip at 154 minutes; 99% of patients same; verified; risk saturates near 0%*	1% to 11%	log-odds 1.64, OR 5.16; range across patients -0.45 to 3.10	4000	train; test agrees (AUC gain 0.081 [0.053, 0.112], n=1000)	no	clearly
C003	direction: state: West Virginia to Washington (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 9%	log-odds 1.66, OR 5.24; range across patients 0.76 to 2.22	122, 72	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	on the training data only
C004	direction: state: West Virginia to California (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 5%	log-odds 1.42, OR 4.15; range across patients 1.42 to 1.42	122, 41	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C005	direction: voice-mail messages: 0 messages to 32 messages (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91; range across patients 1.36 to 1.36	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C006	direction: state: West Virginia to Montana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 4%	log-odds 1.17, OR 3.23; range across patients 1.17 to 1.17	122, 75	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C007	direction: state: West Virginia to Texas (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to 3%	log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96	122, 95	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C008	direction: total evening minutes: 138 minutes to 266 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C009	direction: total international charge: 1.84 US dollars to 3.70 US dollars (whole model) *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.61, OR 1.85; range across patients 0.15 to 0.79	4000	train; test unclear (AUC gain 0.006 [-4e-05, 0.012], n=1000)	yes	slightly; on the training data only
C010	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (whole model) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -3.26, OR 0.04; range across patients -3.26 to -3.26	2939, 1061	train; test agrees (AUC gain 0.024 [0.0005, 0.047], n=1000)	yes	slightly
C011	direction: customer-service calls: 0 calls to 3 calls (whole model) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	slightly
C012	direction: state: West Virginia to Hawaii (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland ...*	1% to <1%	log-odds -1.86, OR 0.16; range across patients -1.86 to -1.86	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C013	direction: state: West Virginia to South Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 66	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C014	direction: state: West Virginia to Maryland (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.60, OR 1.83; range across patients 0.60 to 0.60	122, 82	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C015	direction: total night-time minutes: 136 minutes to 264 minutes (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.70 to 0.70	4000	train; test unclear (AUC gain 0.004 [-0.0008, 0.008], n=1000)	yes	slightly; on the training data only
C016	direction: state: West Virginia to Nevada (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	122, 77	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C017	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (whole model) *shape: dip at 12.96 US dollars; 71% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds -0.20, OR 0.82; range across patients -1.79 to 0.87	4000	train; test unclear (AUC gain 0.005 [-0.005, 0.015], n=1000)	no	slightly; on the training data only
C018	direction: state: West Virginia to Kentucky (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.47, OR 1.59; range across patients 0.47 to 0.47	122, 81	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C019	direction: state: West Virginia to Massachusetts (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.42, OR 1.52; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C020	direction: state: West Virginia to Virginia (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	122, 91	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C021	direction: state: West Virginia to Ohio (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.38, OR 1.46; range across patients 0.38 to 0.38	122, 87	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C022	direction: state: West Virginia to Minnesota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	122, 100	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C023	direction: state: West Virginia to Alaska (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -1.01, OR 0.36; range across patients -2.44 to -0.52	122, 56	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C024	direction: state: West Virginia to Oregon (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 2%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 84	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C025	direction: total international calls: 2 calls to 8 calls (whole model) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.38, OR 0.68; range across patients -0.38 to -0.38	4000	train; test unclear (AUC gain 0.0005 [-0.002, 0.003], n=1000)	yes	slightly; on the training data only
C026	direction: state: West Virginia to South Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.06, OR 1.07; range across patients -0.51 to 0.78	122, 66	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	no	slightly; on the training data only
C027	direction: account length: 48 to 152 (whole model) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C028	direction: state: West Virginia to Missouri (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.17, OR 0.85; range across patients -0.17 to -0.17	122, 80	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C029	direction: state: West Virginia to Washington DC (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.12, OR 0.89; range across patients -0.12 to -0.12	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C030	direction: state: West Virginia to Utah (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.08, OR 1.09; range across patients 0.08 to 0.09	122, 89	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C031	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Kentucky, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Ohio, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Utah, Virginia, Washington, West Virginia, Washington DC, Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.08 to 0.08	122, 1947	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C032	direction: area code: area code 415 to area code 510 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.08, OR 0.92; range across patients -0.08 to -0.08	2016, 960	train; test disagrees (AUC gain -0.0003 [-0.001, 0.0004], n=1000)	yes	slightly; on the training data only
C033	direction: state: West Virginia to North Dakota (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.08, OR 0.93; range across patients -0.08 to -0.08	122, 68	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C034	direction: state: West Virginia to Arizona (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.07, OR 1.07; range across patients 0.07 to 0.07	122, 73	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C035	direction: state: West Virginia to Oklahoma (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 79	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C036	direction: state: West Virginia to North Carolina (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.05, OR 0.95; range across patients -0.05 to -0.05	122, 74	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C037	direction: state: West Virginia to Delaware (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds 0.04, OR 1.04; range across patients 0.04 to 0.04	122, 81	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C038	direction: area code: area code 415 to area code 408 (whole model) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1024	train; test disagrees (AUC gain -0.0003 [-0.001, 0.0004], n=1000)	yes	slightly; on the training data only
C039	direction: state: West Virginia to Indiana (whole model) *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 82	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	slightly; on the training data only
C040	direction: total daytime calls: 75 calls to 125 calls (whole model) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C041	magnitude: international calling plan: no international plan to an international plan, typical patient *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C042	magnitude: state: West Virginia to Washington, typical patient *shape: categorical; highest to lowest: Washington, California, Montana, Texas, South Carolina, Maryland …*	1% to 9%	log-odds 2.08, OR 8.00	122, 72	train; test disagrees (AUC gain -0.002 [-0.012, 0.008], n=1000)	yes	on the training data only
C043	magnitude: voice-mail messages: 0 messages to 32 messages, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C044	magnitude: total evening minutes: 138 minutes to 266 minutes, typical patient *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C045	magnitude: total international charge: 1.84 US dollars to 3.70 US dollars, typical patient *shape: accelerating; 93% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.79, OR 2.20	4000	train; test unclear (AUC gain 0.006 [-4e-05, 0.012], n=1000)	yes	slightly; on the training data only
C046	null: model ignores total daytime charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C047	null: model ignores total evening calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C048	null: model ignores total international minutes (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C049	null: model ignores total night-time calls (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C050	null: model ignores total night-time charge (exact rule)		max change 0.000 (max |change in log-odds|)	4000	train	yes	
C051	structure: made of linear_backbone, service_call_nonlinear, voicemail_chain, intl_state44_unit, area_code_units, state_offsets; added		log-odds n/a, OR n/a	4000	train	yes	
C052	direction: international calling plan: no international plan to an international plan (through this part) *shape: categorical; highest to lowest: an international plan, no international plan*	1% to 14%	log-odds 2.60, OR 13.5; range across patients 2.60 to 2.60	3620, 380	train; test agrees (AUC gain 0.056 [0.032, 0.082], n=1000)	yes	clearly
C053	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 11%	log-odds 4.70, OR 110; range across patients 4.70 to 4.70	4000	train; test agrees (AUC gain 0.157 [0.117, 0.200], n=1000)	yes	clearly
C054	direction: state: West Virginia to Washington (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.47, OR 4.33; range across patients 1.47 to 1.47	122, 72	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C055	direction: state: West Virginia to California (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 5%	log-odds 1.42, OR 4.15; range across patients 1.42 to 1.42	122, 41	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C056	direction: voice-mail messages: 0 messages to 32 messages (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 4%	log-odds 1.36, OR 3.91; range across patients 1.36 to 1.36	4000	train; test agrees (AUC gain 0.018 [0.006, 0.030], n=1000)	yes	slightly
C057	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	<1% to 3%	log-odds 1.73, OR 5.63; range across patients 1.73 to 1.73	4000	train; test agrees (AUC gain 0.020 [0.008, 0.032], n=1000)	yes	slightly
C058	direction: state: West Virginia to Texas (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 3%	log-odds 0.96, OR 2.60; range across patients 0.96 to 0.96	122, 95	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C059	direction: total evening minutes: 138 minutes to 266 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 1.26, OR 3.54; range across patients 1.26 to 1.26	4000	train; test agrees (AUC gain 0.014 [0.004, 0.024], n=1000)	yes	slightly
C060	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.87, OR 2.40; range across patients 0.87 to 0.87	4000	train; test agrees (AUC gain 0.007 [0.0006, 0.014], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C061	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.98, OR 0.14; range across patients -1.98 to -1.98	2939, 1061	train; test disagrees (AUC gain -0.0003 [-0.015, 0.015], n=1000)	yes	slightly; on the training data only
C062	direction: state: West Virginia to Hawaii (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to <1%	log-odds -1.86, OR 0.16; range across patients -1.86 to -1.86	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C063	direction: state: West Virginia to South Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.61, OR 1.84; range across patients 0.61 to 0.61	122, 66	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C064	direction: state: West Virginia to Maryland (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.60, OR 1.83; range across patients 0.60 to 0.60	122, 82	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C065	direction: total night-time minutes: 136 minutes to 264 minutes (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.70, OR 2.01; range across patients 0.70 to 0.70	4000	train; test unclear (AUC gain 0.004 [-0.0008, 0.008], n=1000)	yes	slightly; on the training data only
C066	direction: state: West Virginia to Nevada (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.53, OR 1.70; range across patients 0.53 to 0.53	122, 77	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C067	direction: state: West Virginia to Montana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.46, OR 1.59; range across patients 0.46 to 0.46	122, 75	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C068	direction: state: West Virginia to Massachusetts (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.42, OR 1.52; range across patients 0.42 to 0.42	122, 86	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C069	direction: state: West Virginia to Virginia (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 1%	log-odds -0.63, OR 0.53; range across patients -0.63 to -0.63	122, 91	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C070	direction: state: West Virginia to North Carolina (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.37, OR 1.45; range across patients 0.37 to 0.37	122, 74	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C071	direction: state: West Virginia to Minnesota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 2%	log-odds 0.36, OR 1.44; range across patients 0.36 to 0.36	122, 100	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C072	direction: state: West Virginia to Alaska (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada ...*	1% to 1%	log-odds -0.52, OR 0.59; range across patients -0.52 to -0.52	122, 56	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C073	direction: state: West Virginia to South Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.51, OR 0.60; range across patients -0.51 to -0.51	122, 66	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C074	direction: state: West Virginia to Oregon (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 2%	log-odds 0.31, OR 1.36; range across patients 0.31 to 0.31	122, 84	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C075	direction: total international calls: 2 calls to 8 calls (through this part) *shape: steady (equal steps); 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds -0.38, OR 0.68; range across patients -0.38 to -0.38	4000	train; test unclear (AUC gain 0.0005 [-0.002, 0.003], n=1000)	yes	slightly; on the training data only
C076	direction: account length: 48 to 152 (through this part) *shape: steady; 100% of patients same; verified; risk saturates near 0%*	1% to 1%	log-odds 0.33, OR 1.39; range across patients 0.33 to 0.33	4000	train; test disagrees (AUC gain -0.001 [-0.004, 0.0006], n=1000)	yes	slightly; on the training data only
C077	direction: state: West Virginia to Missouri (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.17, OR 0.85; range across patients -0.17 to -0.17	122, 80	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C078	direction: state: West Virginia to Arizona (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.11, OR 1.12; range across patients 0.11 to 0.11	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C079	direction: state: West Virginia to Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.12, OR 0.89; range across patients -0.12 to -0.12	122, 73	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C080	direction: state: West Virginia to state other than Alaska, Hawaii, Indiana, Massachusetts, Maryland, Minnesota, Missouri, Montana, North Carolina, North Dakota, Arizona, Nevada, Oklahoma, Oregon, California, South Carolina, South Dakota, Texas, Virginia, Washington, West Virginia, Washington DC (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.08, OR 1.08; range across patients 0.08 to 0.08	122, 2285	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C081	direction: state: West Virginia to Oklahoma (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds 0.06, OR 1.06; range across patients 0.06 to 0.06	122, 79	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C082	direction: area code: area code other than area code 510 to area code 510 (through this part) *shape: categorical; highest to lowest: area code other than area code 510, area code 510*	1% to 1%	log-odds -0.06, OR 0.94; range across patients -0.06 to -0.06	3040, 960	train; test disagrees (AUC gain -6e-05 [-0.0006, 0.0005], n=1000)	yes	slightly; on the training data only
C083	direction: state: West Virginia to North Dakota (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.06, OR 0.94; range across patients -0.06 to -0.06	122, 68	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only
C084	direction: state: West Virginia to Indiana (through this part) *shape: categorical; highest to lowest: Washington, California, Texas, South Carolina, Maryland, Nevada …*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	122, 82	train; test disagrees (AUC gain -0.004 [-0.013, 0.004], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C085	structure: its formula reproduces it on 4000 training patients (largest difference 5.32907e-15)		log-odds 5e-15, OR n/a	4000	train	yes	
C086	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	log-odds -3.06, OR 0.05; range across patients -5.15 to -1.60	4000	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C087	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	log-odds -1.11, OR 0.33; range across patients -1.57 to -0.94	4000	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C088	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 94% of patients same; verified; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	slightly
C089	direction: total evening charge: 11.76 US dollars to 22.57 US dollars (through this part) *shape: hinge at 18.08 US dollars; 67% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 12.96 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	2% to 1%	log-odds -1.08, OR 0.34; range across patients -2.67 to 0.00	4000	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C090	direction: state: state other than Alaska, South Dakota, Washington to Washington (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds -0.57, OR 0.57; range across patients -1.46 to 0.00	3806, 72	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C091	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds 0.57, OR 1.77; range across patients 0.00 to 1.29	3806, 66	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C092	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds -0.50, OR 0.61; range across patients -1.92 to 0.00	3806, 56	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C093	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1%	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C094	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Washington minus state other than Alaska, South Dakota, Washington; stronger in Washington *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Washington: 4% to 18%; state other than Alaska, South Dakota, Washington: 1% to 2%	log-odds difference 1.14, OR 3.11	72, 3806	train; test agrees (AUC gain 0.0004 [-0.0005, 0.001], n=1000); group difference on test 1.15 [1.14, 1.17]	yes	clearly
C095	interaction: effect of total daytime minutes (112 minutes to 249 minutes) by total evening charge: total evening charge above 17.12 US dollars minus total evening charge at or below 17.12 US dollars; stronger in	total evening charge above 17.12 US dollars: 1% to 13%; total evening charge at or below 17.12 US dollars: 2% to	log-odds difference 1.47, OR 4.36	1998, 2002	train; test agrees (AUC gain 0.008 [-0.003, 0.019], n=1000); group difference on test 1.49 [1.33, 1.64]	yes	clearly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C097	interaction: effect of total evening charge (11.76 US dollars to 22.57 US dollars) by state: Alaska minus state other than Alaska, South Dakota, Washington; stronger in Alaska *shape: steady; 100% of patients same; verified; risk saturates near 0%*	Alaska: <1% to 1%; state other than Alaska, South Dakota, Washington: 1% to 2%	log-odds difference 1.08, OR 2.95	56, 3806	train; test agrees (AUC gain 0.0004 [-0.0005, 0.001], n=1000); group difference on test 1.14 [1.07, 1.21]	yes	slightly
C098	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C098 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	fold 166 minutes (IQR 151 to 188); past it log-odds -4.16, OR 0.02; range across patients -6.25 to -2.68	3999, 1743	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C099	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 93% of patients same; verified*	3% to 8% (typical patient with a fold)	fold 3.19 calls (IQR 2.36 to 3.19); past it log-odds 1.47, OR 4.35; range across patients 0.52 to 3.17	2700, 528	train; test agrees (AUC gain 0.095 [0.059, 0.134], n=1000)	yes	clearly
C100	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C100 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 13.7 to 18.8); past it log-odds -1.95, OR 0.14; range across patients -3.27 to 0.00	71, 35	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C101	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C101 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.47); past it log-odds -1.47, OR 0.23; range across patients -2.17 to -1.19	3886, 2729	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C102	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C102 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	fold 75.0 calls (IQR 68.2 to 96.7); past it log-odds 0.31, OR 1.37; range across patients 0.02 to 0.45	12, 9 (small n)	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; the model has little data on this; on the training data only
C103	structure: made of csc_usage_gate, csc_hinge; added		log-odds n/a, OR n/a	4000	train	yes	
C104	direction: total daytime minutes: 112 minutes to 249 minutes (through this part) *shape: hinge at 166 minutes; 71% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	12% to 1%	log-odds -3.06, OR 0.05; range across patients -5.15 to -1.60	4000	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C105	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: hinge at 2.95 US dollars; 88% of patients same; fallback; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	3% to 1%	log-odds -1.11, OR 0.33; range across patients -1.57 to -0.94	4000	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C106	direction: customer-service calls: 0 calls to 3 calls (through this part) *shape: accelerating; 77% of patients same; fallback; risk saturates near 0%*	1% to 2%	log-odds 0.94, OR 2.56; range across patients 0.00 to 2.43	4000	train; test agrees (AUC gain 0.062 [0.036, 0.092], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C109	direction: state: state other than Alaska, South Dakota, Washington to South Dakota (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1%	log-odds 0.57, OR 1.77; range across patients 0.00 to 1.29	3806, 66	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C110	direction: state: state other than Alaska, South Dakota, Washington to Alaska (through this part) *shape: categorical; highest to lowest: South Dakota, Alaska, state other than Alaska, South Dakota, Washington, Washington*	1% to 1% (typical patient this part acts on)	log-odds -0.50, OR 0.61; range across patients -1.92 to 0.00	3806, 56	train; test unclear (AUC gain 0.0004 [-0.0005, 0.001], n=1000)	yes	slightly; on the training data only
C111	direction: total daytime calls: 75 calls to 125 calls (through this part) *shape: steady; 87% of patients same; fallback; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds 0.13, OR 1.14; range across patients 0.00 to 0.31	4000	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; on the training data only
C112	threshold: total daytime minutes: fold at about 166 minutes (stops) *shape: hinge at 166 minutes; 72% of patients same; fallback; risk saturates near 0%; one point with C112 (median fold 166 minutes, typical patient 160 minutes, 10th-90th percentile 145 minutes to 213 minutes, wide)*	28% to 1% (typical patient with a fold)	fold 166 minutes (IQR 151 to 188); past it log-odds -4.16, OR 0.02; range across patients -6.25 to -2.68	3999, 1743	train; test agrees (AUC gain 0.059 [0.034, 0.089], n=1000)	yes	clearly
C113	threshold: total evening charge: fold at about 18.08 US dollars (stops) for customers in Colorado *shape: hinge at 18.08 US dollars; 68% of patients same; fallback; risk saturates near 0%; one point with C113 (median fold 18.08 US dollars, typical patient 16.50 US dollars, 10th-90th percentile 7.14 US dollars to 21.50 US dollars, wide)*	4% to 1% (typical patient with a fold (customers in Colorado))	fold 18.1 US dollars (IQR 13.7 to 18.8); past it log-odds -1.95, OR 0.14; range across patients -3.27 to 0.00	71, 35	train; test unclear (AUC gain 0.008 [-0.003, 0.019], n=1000)	yes	slightly; on the training data only
C114	threshold: customer-service calls: fold at about 4 calls (starts) for customers in Wisconsin *shape: accelerating; 61% of patients same; fallback; risk saturates near 0%*	2% to 5% (typical patient with a fold (customers in Wisconsin))	fold 3.08 calls (IQR 2.33 to 4.11); past it log-odds 1.11, OR 3.04; range across patients 0.00 to 2.88	59, 12 (small n)	train; test agrees (AUC gain 0.062 [0.036, 0.092], n=1000)	yes	slightly; the model has little data on this
C115	threshold: total international charge: fold at about 2.95 US dollars (stops) *shape: hinge at 2.95 US dollars; 93% of patients same; verified; risk saturates near 0%; one point with C115 (median fold 2.95 US dollars, typical patient 2.92 US dollars, 10th-90th percentile 2.95 US dollars to 4.10 US dollars, wide)*	4% to 1% (typical patient with a fold)	fold 2.95 US dollars (IQR 2.95 to 3.47); past it log-odds -1.47, OR 0.23; range across patients -2.17 to -1.19	3886, 2729	train; test unclear (AUC gain 0.004 [-0.004, 0.012], n=1000)	yes	slightly; on the training data only
C116	threshold: total daytime calls: fold at about 75 calls (starts) for customers in Louisiana *shape: hinge at 93 calls; 80% of patients same; fallback; risk saturates near 0%; one point with C116 (median fold 75 calls, typical patient 93 calls, 10th-90th percentile 37 calls to 139 calls, wide)*	1% to 1% (typical patient with a fold (customers in Louisiana))	fold 75.0 calls (IQR 68.2 to 96.7); past it log-odds 0.31, OR 1.37; range across patients 0.02 to 0.45	12, 9 (small n)	train; test disagrees (AUC gain -0.001 [-0.003, 0.0005], n=1000)	yes	slightly; the model has little data on this; on the training data only
C117	subgroup: who this part acts on: 3063 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.77; log-odds 1.77, OR 5.84; range across patients 0.17 to 4.40	3063, 937	train; test agrees (AUC gain 0.134 [0.089, 0.181], n=1000)	yes	slightly
C118	structure: its formula reproduces it on 4000 training patients (largest difference 1.33227e-15)		log-odds 1e-15, OR n/a	4000	train	yes	
C119	direction: voice-mail plan: no voice-mail plan to a voice-mail plan (through this part) *shape: categorical; highest to lowest: no voice-mail plan, a voice-mail plan*	1% to <1%	log-odds -1.29, OR 0.28; range across patients -1.29 to -1.29	2939, 1061	train; test disagrees (AUC gain -0.005 [-0.015, 0.005], n=1000)	yes	slightly; on the training data only
C120	direction: state: state other than North Carolina to North Carolina (through this part) *shape: categorical; highest*	1% to 1%	log-odds -0.42, OR 0.66; range across patients -0.42 to -0.42	3926, 74	train; test agrees (AUC gain 0.0004 [0.0001, 0.0008], n=1000)	yes	slightly

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C121	structure: its formula reproduces it on 4000 training patients (largest difference 2.22045e-16)		log-odds 2e-16, OR n/a	4000	train	yes	
C122	threshold: customer-service calls: fold at about 4 calls (starts) *shape: accelerating; 100% of patients same; verified; risk saturates near 0%*	1% to 2% (typical patient with a fold)	fold 3.19 calls (IQR 3.19 to 3.19); past it log-odds 0.52, OR 1.69; range across patients 0.52 to 0.52	4000, 318	train; test agrees (AUC gain 0.006 [0.0008, 0.012], n=1000)	yes	slightly
C123	subgroup: who this part acts on: 318 of 4000 patients	7% to 12% (typical patient this part acts on (risk without the part, then with it))	share 0.08; log-odds 0.89, OR 2.44; range across patients 0.52 to 1.82	318, 3682	train; test agrees (AUC gain 0.006 [0.0008, 0.012], n=1000)	yes	slightly
C124	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C125	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Washington (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.76, OR 2.14; range across patients 0.76 to 0.76	3386, 72	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C126	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Montana (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.71, OR 2.04; range across patients 0.71 to 0.71	3386, 75	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C127	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Kentucky (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.39, OR 1.47; range across patients 0.39 to 0.39	3386, 81	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C128	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Ohio (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 2%	log-odds 0.30, OR 1.35; range across patients 0.30 to 0.30	3386, 87	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C129	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Arizona (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware, Nevada …*	1% to 1%	log-odds -0.05, OR 0.96; range across patients -0.05 to -0.05	3386, 73	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only
C130	direction: state: state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio, Washington, Delaware to Delaware (through this part) *shape: categorical; highest to lowest: Washington, Montana, Kentucky, Ohio, state other than Kentucky, Montana, North Dakota, Arizona, Nevada, Ohio*	1% to 1%	log-odds -0.04, OR 0.96; range across patients -0.04 to -0.04	3386, 81	train; test unclear (AUC gain 0.0005 [-0.004, 0.005], n=1000)	yes	slightly; on the training data only

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C133	structure: its formula reproduces it on 4000 training patients (largest difference 1.77636e-15)		log-odds 2e-15, OR n/a	4000	train	yes	
C134	direction: area code: area code 415 to area code 408 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 1024	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	yes	slightly; on the training data only
C135	direction: area code: area code 415 to area code 510 (through this part) *shape: categorical; highest to lowest: area code 415, area code 408, area code 510*	1% to 1%	log-odds -0.02, OR 0.98; range across patients -0.02 to -0.02	2016, 960	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	yes	slightly; on the training data only
C136	subgroup: who this part acts on: 2016 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.50; log-odds 0.02, OR 1.02; range across patients 0.02 to 0.02	2016, 1984	train; test disagrees (AUC gain -0.0002 [-0.0005, 6e-05], n=1000)	no	slightly; on the training data only
C137	null: this part barely changes predictions (switched off)		max change 0.005 (max |change in predicted risk|)	4000	train	yes	
C138	structure: its formula reproduces it on 4000 training patients (largest difference 0.0)		log-odds 0.00, OR n/a	4000	train	yes	
C139	direction: total international charge: 1.84 US dollars to 3.70 US dollars (through this part) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient this part acts on)	log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	yes	slightly; on the training data only
C140	direction: state: state other than Utah to Utah (through this part) *shape: categorical; highest to lowest: Utah, state other than Utah*	1% to 1%	log-odds 0.005, OR 1.01; range across patients 0.00 to 0.01	3911, 89	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	yes	slightly; on the training data only
C141	interaction: effect of total international charge (1.84 US dollars to 3.70 US dollars) by state: Utah minus state other than Utah; stronger in Utah *shape: accelerating; 96% of patients same; verified; risk saturates near 0%*	Utah: 1% to 3%; state other than Utah: 1% to 2%	log-odds difference 0.003, OR 1.00	89, 3911	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000); group difference on test 0.03 [-0.09, 0.12]	no	slightly; on the training data only
C142	threshold: total international charge: fold at about 0.77 US dollars (stops) *shape: flat; 100% of patients same; verified; risk saturates near 0%*	1% to 1% (typical patient with a fold)	fold 0.77 US dollars (IQR 0.77 to 0.77); past it log-odds -0.0003, OR 1.00; range across patients 0.00 to 0.00	4000, 100	train; test unclear (AUC gain 8e-06 [0.000, 5e-05], n=1000)	no	slightly; on the training data only
C143	subgroup: who this part acts on: 100 of 4000 patients	1% to 1% (typical patient this part acts on (risk without the part, then with it))	share 0.03; log-odds 0.007, OR 1.01; range across patients 0.001 to 0.01	100, 3900	train; test unclear (AUC gain 0.000 [0.000, 0.000], n=1000)	no	slightly; on the training data only
C144	null: this part barely changes predictions (switched off)		max change 0.002 (max |change in predicted risk|)	4000	train	yes	

Claim	What is checked	Risk, typical patient	Estimate (log-odds; OR); range across patients	n	Test check	Verified	Words
C145	structure: its formula reproduces it on 4000 training patients (largest difference 4.44089e-16)		log-odds 4e-16, OR n/a	4000	train	yes	
C146	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	
C147	boundary: boundary		log-odds n/a, OR n/a	4000	train	no	

Not supported by the model.

- C002: direction claim about total daytime minutes in the whole model: not supported by the model (14% of patients have the opposite sign (allowed: 10%)).
- C017: direction claim about total evening charge in the whole model: not supported by the model (52% of patients have the opposite sign (allowed: 10%)).
- C026: direction claim about state in the whole model: not supported by the model (53% of patients have the opposite sign (allowed: 10%)).

- C136: subgroup claim about area code in part ann_94: not supported by the model (its effect is negligible).
- C141: interaction claim about total international charge and state in part ann_93: not supported by the model (the group difference is below 0.1 log-odds; the part's own counterfactual difference has the opposite sign).
- C142: threshold claim about total international charge in part ann_93: not supported by the model (the fold changes the curve by only 0.037 log-odds compared with a straight line (needed 0.1); the effect past the fold is too small or not consistent across patients).
- C143: subgroup claim about state and total international charge in part ann_93: not supported by the model (its effect is negligible).
- C146: boundary claim about number of calls to customer service and total daytime minutes in part ann_96: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).
- C147: boundary claim about total daytime minutes and number of calls to customer service in part ann_90: not supported by the model (no switch-on line could be traced on the first map to check against the data; no switch-on boundary in the observed range).

1.2 Coverage

Coverage is computed by the current code (the function behind the `get_coverage` tool) under these conventions. *Candidate nodes* are all nodes of the explained model state except the output nodes. A candidate node is *covered* when some annotation of the hierarchy (a leaf, or a composition together with its descendants) contains it and all its enabled outgoing edges. *Exact Null* parts, which have no effect on any output for any input, count as covered: nodes with no path to an output over enabled, non-zero-weight edges (disconnected inputs and dead ends), and edges that have zero weight or touch such a node. Compositional coverage is the fraction of required compositions present: every annotation except a single global root containing every output must have a parent; a parentless, childless annotation made only of Null nodes is exempt.

- **Null, disconnected input (30):** `-11` (state:18), `-13` (state:2), `-15` (state:21), `-16` (state:22), `-19` (state:25), `-2` (state:1), `-23` (state:29), `-25` (state:30), `-26` (state:31), `-27` (state:32), `-29` (state:34), `-3` (state:10), `-33` (state:38), `-34` (state:39), `-38` (state:42), `-42` (state:46), `-44` (state:48), `-46` (state:5), `-47` (state:50), `-48` (state:6), `-5` (state:12), `-51` (state:9), `-6` (state:13), `-61` (totalDayCharge), `-63` (totalEveCalls), `-66` (totalNightCalls), `-67` (totalNightCharge), `-68` (totalIntlMinutes), `-7` (state:14), `-9` (state:16).
- **Uncovered nodes:** none.
- **Uncovered edges:** none.
- **Missing compositions:** none.

1.3 Annotation hierarchy

The hierarchy has 1 root; the global root is churn_logit. Tags (A1, A2, ...) are used in the figures.

▷ A1: churn_logit (composition, section 1.6.1)
- ▷ A2: linear_backbone (leaf, section 1.6.2)
- ▷ A3: voicemail_chain (leaf, section 1.6.3)
- ▷ A4: intl_state44_unit (leaf, section 1.6.4)
- ▷ A5: area_code_units (leaf, section 1.6.5)
- ▷ A6: state_offsets (leaf, section 1.6.6)
- ▷ A7: service_call_nonlinear (composition, section 1.6.7)
  - ▷ A8: csc_usage_gate (leaf, section 1.6.8)
  - ▷ A9: csc_hinge (leaf, section 1.6.9)

1.4 Network figures

The explained model state has more than 60 non-Null nodes, so the collapsed views are shown; the full view is in `churn100-s2-c2-fig-full.tex`.

1.5 Refactoring operations

The explained model state is the phenotype after these operations, replayed in order. All are function-preserving; the largest change in the model output over the 4000 training rows is 5.55×10^{-16} (tolerance 1.00×10^{-9}).

- #71: split `-71` into `-71_a`, `-71_b`, `-71_c`, `-71_d`, `-71_e`.
- #72: split `-70` into `-70_a`, `-70_b`, `-70_c`.
- #73: split `-64` into `-64_a`, `-64_b`.
- #74: split `-59` into `-59_a`, `-59_b`.
- #75: split `-57` into `-57_a`, `-57_b`, `-57_c`, `-57_d`.
- #76: split `-43` into `-43_a`, `-43_b`, `-43_c`.
- #77: split `-37` into `-37_a`, `-37_b`.
- #78: split `-28` into `-28_a`, `-28_b`.
- #79: split `-24` into `-24_a`, `-24_b`.
- #80: split `-22` into `-22_a`, `-22_b`.
- #81: split `-21` into `-21_a`, `-21_b`.
- #82: split `-20` into `-20_a`, `-20_b`, `-20_c`.
- #83: split `-1` into `-1_a`, `-1_b`.
- #84: consolidate `-71_a`, `-71_b`, `-71_d`, `-71_e`.
- #85: consolidate `-57_b`, `-57_c`, `-57_d`.
- #86: consolidate `-20_b`, `-20_c`.
- #87: identity node `identity_lin` before `0` (33 connection(s)).
- #88: identity node `identity_svc` before `0` (3 connection(s)).

1.6 Annotations

1.6.1 A1: churn_logit

Kind: Composition annotation (depth 0); children A2 (section 1.6.2), A3 (section 1.6.3), A4 (section 1.6.4), A5 (section 1.6.5), A6 (section 1.6.6), A7 (section 1.6.7).

- **Entry nodes:** `-69` (totalIntlCalls), `-68` (totalIntlMinutes), `-67` (totalNightCharge), `-66` (totalNightCalls), `-65` (totalNightMinutes), `-63` (totalEveCalls), `-62` (totalEveMinutes), `-61` (totalDayCharge), `-60` (totalDayCalls), `-58` (numberVmailMessages), `-56` (internationalPlan), `-55` (areaCode:510), `-54` (areaCode:415), `-53` (areaCode:408), `-52` (accountLength), `-51` (state:9), `-50` (state:8), `-49` (state:7), `-48` (state:6), `-47` (state:50), `-46` (state:5), `-45` (state:49), `-44` (state:48), `-42` (state:46), `-41` (state:45), `-40` (state:44), `-39` (state:43), `-38` (state:42), `-36` (state:40), `-35` (state:4), `-34` (state:39), `-33` (state:38), `-32` (state:37), `-31` (state:36),

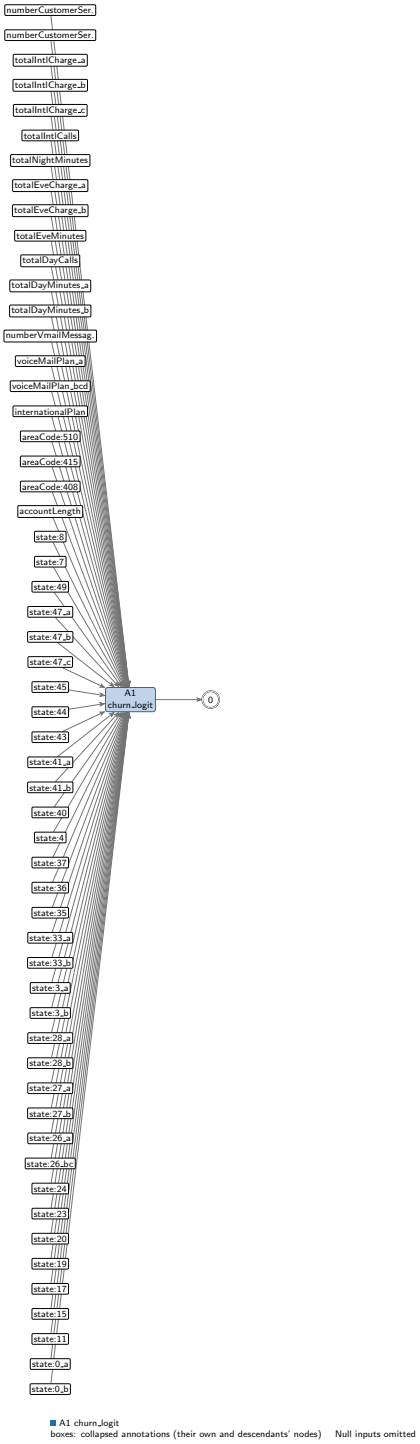


Figure 1: churn100-s2-c2: collapsed view at hierarchy level 0 (fully recomposed).

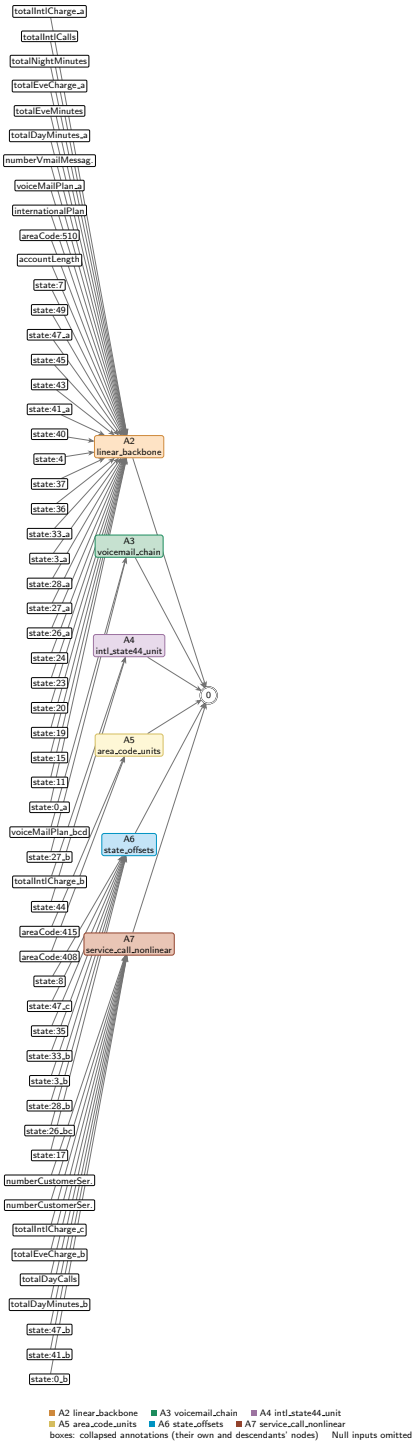


Figure 2: churn100-s2-c2: collapsed view at hierarchy level 1.

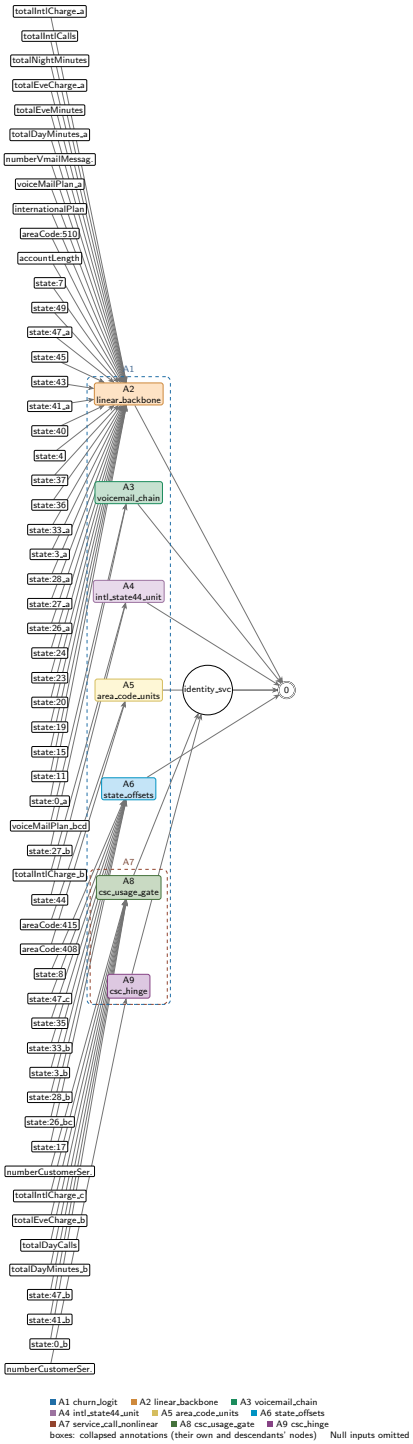


Figure 3: churn100-s2-c2: collapsed view at hierarchy level 2.

-30 (state:35), -29 (state:34), -27 (state:32), -26 (state:31), -25 (state:30), -23 (state:29), -19 (state:25), -18 (state:24), -17 (state:23), -16 (state:22), -15 (state:21), -14 (state:20), -13 (state:2), -12 (state:19), -11 (state:18), -10 (state:17), -9 (state:16), -8 (state:15), -7 (state:14), -6 (state:13), -5 (state:12), -4 (state:11), -3 (state:10), -2 (state:1), -71_c (numberCustomerServiceCalls_c), -70_a (totalIntlCharge_a), ... (26 more; see the JSON record).

- **Exit nodes:** 0.
- **Internal nodes:** none.
- **Edges (weight):** 161→0 (0.0489), 252→0 (0.00702), 1133→0 (-0.891), 4051→0 (-0.999), 7647→0 (-0.00656), 8418→0 (-0.00263), 8896→0 (-0.00632), 10332→0 (0.115), 13551→0 (0.0461), 13725→0 (0.0479), 14653→0 (-0.254), 16344→0 (0.0539), 16902→0 (0.0548), identity_lin→0 (1), identity_svc→0 (1).
- **With its descendants:** 114 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 113 candidate nodes.

Hypothesis. Root: logit P(churn) = -1.15 + linear_backbone + service_call_nonlinear (identity_svc) - 0.89*1133 - 1.00*4051 (voicemail_chain) + state_offsets + 0.05*161 - 0.25*14653 (area codes, 14653 dead) + 0.007*252, purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, night charge, night calls, evening calls, day charge and 25 state indicators) and have zero effect. Train AUC 0.8944, accuracy 0.896. Performance sources (train): all children replaced by least-squares linear fits gives AUC 0.8314 (an L2 logistic regression refit on all 71 inputs gets 0.8433); restoring csc_usage_gate alone gives 0.8917, csc_hinge alone 0.8462, the others 0.8314 (exactly or nearly linear). So the gain over linear comes from the service-call x low-usage gate. Rare-category switches: state:11 -1.93 logit (73 rows, empirical -1.35); state:47 +1.47 mean (72 rows, empirical +0.92; direct +1.39, +0.76 via 10332, but it switches off 1264, down to -1.38 for some); state:4 +1.34 (41 rows, +0.80); state:0 -1.13 mean, down to -5.0 (56 rows, -0.76; switches off 2796 in the gate); state:26 +1.09 (75 rows, +0.50; via direct + state offsets); state:43 +0.88 (95 rows, +0.48); state:45 -0.71 (91 rows, -0.85).

Closed form (children as named functions):

$$
\begin{aligned}
y_0 &= \sigma(z_0) \\
z_0 &= -1.15 + 1.0\,\text{linear_backbone}\big(x_{state:19}, x_{state:20}, \\
&\quad x_{state:23}, x_{state:24}, x_{state:0_a}, x_{state:26_a}, x_{state:27_a}, x_{state:28_a}, \\
&\quad x_{state:3_a}, x_{state:33_a}, x_{state:36}, x_{state:37}, x_{state:4}, x_{state:40}, \\
&\quad x_{state:41_a}, x_{state:43}, x_{state:11}, x_{state:45}, x_{state:47_a}, x_{state:49}, \\
&\quad x_{state:7}, x_{accountLength}, x_{areaCode:510}, x_{internationalPlan}, x_{voiceMailPlan_a}, \\
&\quad x_{numberVmailMessages}, x_{totalDayMinutes_a}, x_{totalEveMinutes}, x_{totalEveCharge_a}, \\
&\quad x_{totalNightMinutes}, x_{totalIntlCalls}, x_{totalIntlCharge_a}, x_{state:15}\big) \\
&\quad + 1.0\,\text{service_call_nonlinear}\big(x_{numberCustomerServiceCalls_abde}, \\
&\quad x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b}, \\
&\quad x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}, \\
&\quad x_{numberCustomerServiceCalls_c}\big) \\
&\quad + 0.0479\,\text{state_offsets_5}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big) \\
&\quad + 0.0489\,\text{area_code_units_0}\big(x_{areaCode:415}, x_{areaCode:408}\big) \\
&\quad + 0.0539\,\text{state_offsets_6}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big) \\
&\quad + 0.0461\,\text{state_offsets_4}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b}, \\
&\quad x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)
\end{aligned}
$$

$$
\begin{aligned}
&+ 0.0548\,\text{state_offsets_7}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&+ 0.00702\,\text{intl_state44_unit}\big(x_{state:44}, x_{totalIntlCharge_b}\big)\\
&+ 0.115\,\text{state_offsets_3}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.891\,\text{voicemail_chain_0}\big(x_{voiceMailPlan_bcd}, x_{state:27_b}\big)\\
&- 0.00632\,\text{state_offsets_2}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.254\,\text{area_code_units_1}\big(x_{areaCode:415}, x_{areaCode:408}\big)\\
&- 0.00263\,\text{state_offsets_1}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.00656\,\text{state_offsets_0}\big(x_{state:8}, x_{state:28_b}, x_{state:3_b},\\
&x_{state:47_c}, x_{state:35}, x_{state:26_bc}, x_{state:17}, x_{state:33_b}\big)\\
&- 0.999\,\text{voicemail_chain_1}\big(x_{voiceMailPlan_bcd}, x_{state:27_b}\big)
\end{aligned}
$$

The fully expanded form is in the JSON record. where y_0 is node `0`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00617, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Root: logit P(churn) = -1.15 + linear_backbone + service_call_nonlinear (identity_svc) - 0.89*1133 - 1.00*4051 (voicemail_chain) + state_offsets + 0.05*161 - 0.25*14653 (area codes, 14653 dead) + 0.007*252, purely additive in the logit; the sigmoid is the only coupling. 30 inputs are unconnected (international minutes, night charge, night calls, evening calls, day charge and 25 state indicators) a

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.2 A2: linear_backbone

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-12` (state:19), `-14` (state:20), `-17` (state:23), `-18` (state:24), `-1_a` (state:0_a), `-20_a` (state:26_a), `-21_a` (state:27_a), `-22_a` (state:28_a), `-24_a` (state:3_a), `-28_a` (state:33_a), `-31` (state:36), `-32` (state:37), `-35` (state:4), `-36` (state:40), `-37_a` (state:41_a), `-39` (state:43), `-4` (state:11), `-41` (state:45), `-43_a` (state:47_a), `-45` (state:49), `-49` (state:7), `-52` (accountLength), `-55` (areaCode:510), `-56` (internationalPlan), `-57_a` (voiceMailPlan_a), `-58` (numberVmailMessages), `-59_a` (totalDayMinutes_a), `-62` (totalEveMinutes), `-64_a` (totalEveCharge_a), `-65` (totalNightMinutes), `-69` (totalIntlCalls), `-70_a` (totalIntlCharge_a), `-8` (state:15).
- **Exit nodes:** `identity_lin`.
- **Internal nodes:** none.
- **Edges (weight):** `-4`→`identity_lin` (-0.259), `-8`→`identity_lin` (-0.0136), `-12`→`identity_lin` (0.0494), `-14`→`identity_lin` (0.0744), `-17`→`identity_lin` (0.0442), `-18`→`identity_lin` (-0.0346), `-31`→`identity_lin` (-0.00299), `-32`→`identity_lin` (0.0331), `-35`→`identity_lin` (0.135), `-36`→`identity_lin` (0.0673), `-39`→`identity_lin` (0.133), `-41`→`identity_lin` (-0.106), `-45`→`identity_lin` (-0.0136), `-49`→`identity_lin` (-0.0269), `-52`→`identity_lin` (0.126), `-55`→`identity_lin` (-0.0249), `-56`→`identity_lin` (0.763), `-58`→`identity_lin` (0.575), `-62`→`identity_`

lin (0.497), -65→identity_lin (0.274), -69→identity_lin (-0.159), -70_a→identity_lin (0.694), -64_a→identity_lin (0.346), -59_a→identity_lin (1.84), -57_a→identity_lin (-0.872), -43_a→identity_lin (0.184), -37_a→identity_lin (-0.0745), -28_a→identity_lin (0.0622), -24_a→identity_lin (0.00424), -22_a→identity_lin (-0.0176), -21_a→identity_lin (0.0391), -20_a→identity_lin (0.0518), -1_a→identity_lin (-0.0703).

- **Covers** (coverage, own and as part of the hierarchy): 33 candidate nodes.

Hypothesis. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 33 inputs: day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, evening charge +0.35, night minutes +0.27, state:11 -0.26 (-1.93 logit for its 73 customers), state:47 +0.18 (+1.39), state:4 +0.14 (+1.34), state:43 +0.13 (+0.88), account length +0.13, international calls -0.16, state:45 -0.11 (-0.71), and smaller state/area-code terms. Hypothesis: the additive backbone; day minutes, voicemail plan and international plan dominate. Evidence: closed form exact (identity); train sensitivities equal the weights (day minutes 1.84, voicemail plan 0.87, international plan 0.76).

Closed form:

$$
\begin{aligned}
y_0 = {} & 0.0518x_{state:26_a} + 1.84x_{totalDayMinutes_a} + 0.0673x_{state:40} \\
& + 0.346x_{totalEveCharge_a} + 0.0442x_{state:23} + 0.497x_{totalEveMinutes} \\
& + 0.0622x_{state:33_a} + 0.126x_{accountLength} + 0.0331x_{state:37} + 0.133x_{state:43} \\
& + 0.135x_{state:4} + 0.00424x_{state:3_a} + 0.273x_{totalNightMinutes} \\
& + 0.575x_{numberVmailMessages} + 0.0744x_{state:20} + 0.0391x_{state:27_a} \\
& + 0.694x_{totalIntlCharge_a} + 0.184x_{state:47_a} + 0.763x_{internationalPlan} \\
& + 0.0494x_{state:19} - 0.0703x_{state:0_a} - 0.159x_{totalIntlCalls} \\
& - 0.00299x_{state:36} - 0.0136x_{state:49} - 0.0745x_{state:41_a} - 0.0346x_{state:24} \\
& - 0.0249x_{areaCode:510} - 0.106x_{state:45} - 0.0136x_{state:15} - 0.259x_{state:11} \\
& - 0.0176x_{state:28_a} - 0.0269x_{state:7} - 0.872x_{voiceMailPlan_a}
\end{aligned}
$$

where y_0 is node identity_lin; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0165, and the exact function by at most 2.66×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, analytical_formula; data split: train. Closed form from get_formula (force).
- EMPIRICAL, empirical_viz; data split: train. Direct input-to-output edges collected through identity_lin (an identity node). Exactly linear in 33 inputs: day minutes +1.84 per SD, voicemail plan -0.87, international plan +0.76, international charge +0.69, voicemail messages +0.58, evening minutes +0.50, evening charge +0.35, night minutes +0.27, state:11 -0.26 (-1.93 logit for its 73 customers), state:47 +0.18 (+1.39), state:4 +0.14 (+1.34),

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.3 A3: voicemail_chain

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** -57_bcd (voiceMailPlan_bcd), -21_b (state:27_b).
- **Exit nodes:** 1133, 4051.
- **Internal nodes:** 5577, 13852, 14955.
- **Edges (weight):** 1133→4051 (0.942), 5577→1133 (0.191), 13852→1133 (0.221), 14955→1133 (0.0312), -21_b→14955 (0.993), -57_bcd→1133 (0.111), -57_bcd→5577 (0.962), -57_bcd→13852 (1.01).

- **Covers** (coverage, own and as part of the hierarchy): 5 candidate nodes.

Hypothesis. 5577 = relu(0.99 + 0.96 vmPlan), 13852 = relu(-1.52 + 1.01 vmPlan) (on exactly for the 26.5% with a voicemail plan), 14955 = relu(0.51 + 0.99 state:27), 1133 = relu(0.72 + 0.11 vmPlan + 0.19*5577 + 0.22*13852 + 0.03*14955), 4051 = relu(2.09 + 0.94*1133). All but 13852 are always active, and voicemail plan and state:27 are binary, so the chain is exactly affine in them (least-squares R2 = 1.0). Hypothesis: a further voicemail-plan penalty (chain logit -4.76 with a plan vs -3.47 without, i.e. -1.29 on top of the direct -1.97 for having a plan) plus a small state:27 offset. Evidence: exact linear fit; linear replacement AUC 0.8944 (unchanged). Mean ablation RAISES train AUC to 0.8979: the chain's extra voicemail penalty slightly hurts ranking.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&0.723 + 0.111 x_{voiceMailPlan_bcd} \\
&+ 0.191 \operatorname{relu}\big(0.994 + 0.962 x_{voiceMailPlan_bcd}\big) \\
&+ 0.221 \operatorname{relu}\big(-1.52 + 1.01 x_{voiceMailPlan_bcd}\big) \\
&+ 0.0312 \operatorname{relu}\big(0.513 + 0.993 x_{state:27_b}\big)\big) \\
y_1 = \operatorname{relu}\big(&2.09 + 0.942 \operatorname{relu}\big(0.723 \\
&+ 0.111 x_{voiceMailPlan_bcd} + 0.191 \operatorname{relu}\big(0.994 \\
&+ 0.962 x_{voiceMailPlan_bcd}\big) + 0.221 \operatorname{relu}\big(-1.52 \\
&+ 1.01 x_{voiceMailPlan_bcd}\big) + 0.0312 \operatorname{relu}\big(0.513 \\
&+ 0.993 x_{state:27_b}\big)\big)\big)
\end{aligned}
$$

where y_0 is node `1133`; y_1 is node `4051`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00153, and the exact function by at most 4.44×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 5577 = relu(0.99 + 0.96 vmPlan), 13852 = relu(-1.52 + 1.01 vmPlan) (on exactly for the 26.5% with a voicemail plan), 14955 = relu(0.51 + 0.99 state:27), 1133 = relu(0.72 + 0.11 vmPlan + 0.19*5577 + 0.22*13852 + 0.03*14955), 4051 = relu(2.09 + 0.94*1133). All but 13852 are always active, and voicemail plan and state:27 are binary, so the chain is exactly affine in them (least-squares R2 = 1.0). Hyp

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.4 A4: intl_state44_unit

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-40` (state:44), `-70_b` (totalIntlCharge_b).
- **Exit nodes:** `252`.
- **Internal nodes:** none.
- **Edges (weight):** `-40`→`252` (0.432), `-70_b`→`252` (-0.851).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 252 = relu(-2.22 + 0.43 state:44 - 0.85 intlCharge), logit weight +0.007. Active on 100 of 4000 training rows (71 in state:44, the rest with very low international charge); logit contribution at most +0.03. Hypothesis: a negligible state:44 / low-international-charge offset. Evidence: sensitivity 0.01/0.02; mean ablation and linear replacement leave train AUC at 0.8944.

Closed form:

$$y_0 = \mathrm{relu}\big(-2.22 + 0.432 x_{state:44} - 0.851 x_{totalIntlCharge_b}\big)$$

where y_0 is node `252`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.008, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 252 = relu(-2.22 + 0.43 state:44 - 0.85 intlCharge), logit weight +0.007. Active on 100 of 4000 training rows (71 in state:44, the rest with very low international charge); logit contribution at most +0.03. Hypothesis: a negligible state:44 / low-international-charge offset. Evidence: sensitivity 0.01/0.02; mean ablation and linear replacement leave train AUC at 0.8944.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.5 A5: area_code_units

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-54` (areaCode:415), `-53` (areaCode:408).
- **Exit nodes:** `161`, `14653`.
- **Internal nodes:** none.
- **Edges (weight):** `-54`→`161` (0.577), `-53`→`14653` (-0.535).
- **Covers** (coverage, own and as part of the hierarchy): 2 candidate nodes.

Hypothesis. 161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415). 14653 = relu(-1.07 - 0.54 areaCode:408) is never active on train (dead). Hypothesis: area code barely matters (direct area code 510 term -0.02 per SD in the backbone). Evidence: sensitivity 0.29 for 161 on its unit scale, 0 for 14653; linear R2 1.0; mean ablation AUC 0.8945.

Closed form:

$$y_0 = \mathrm{relu}\big(-0.137 + 0.577 x_{areaCode:415}\big)$$
$$y_1 = \mathrm{relu}\big(-1.07 - 0.535 x_{areaCode:408}\big)$$

where y_0 is node `161`; y_1 is node `14653`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 3.38×10^{-4}, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 161 = relu(-0.14 + 0.58 areaCode:415), logit weight +0.05: on exactly for area code 415 (binary), so exactly affine (+0.02 logit for area code 415). 14653 = relu(-1.07 - 0.54 areaCode:408) is never active on train (dead). Hypothesis: area code barely matters (direct area code 510 term -0.02 per SD in the backbone). Evidence: sensitivity 0.29 for 161 on its unit scale, 0 for 14653; linear R2 1.0; m

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.6 A6: state_offsets

Kind: Leaf annotation (depth 1); parent A1 (section 1.6.1).

- **Entry nodes:** `-50` (state:8), `-22_b` (state:28_b), `-24_b` (state:3_b), `-43_c` (state:47_c), `-30` (state:35), `-20_bc` (state:26_bc), `-10` (state:17), `-28_b` (state:33_b).
- **Exit nodes:** `7647`, `8418`, `8896`, `10332`, `13551`, `13725`, `16344`, `16902`.
- **Internal nodes:** `4967`, `11261`.
- **Edges (weight):** `-50`→`4967` (0.979), `-10`→`11261` (0.99), `-30`→`13551` (0.998), `4967`→`7647` (0.984), `11261`→`16902` (1), `-43_c`→`10332` (0.894), `-28_b`→`11261` (-0.0231), `-24_b`→`8896` (1.01), `-22_b`→`8418` (0.987), `-20_bc`→`13725` (0.991), `-20_bc`→`16344` (0.997).
- **Covers** (coverage, own and as part of the hierarchy): 10 candidate nodes.

Hypothesis. Units 4967 -> 7647 (state:8), 8418 (state:28), 8896 (state:3), 10332 (state:47), 13551 (state:35), 13725 and 16344 (state:26), 11261 -> 16902 (state:17, state:33). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest offsets here: state:47 +0.76 (10332, adding to its direct +1.39), state:26 +0.71 (13725 + 16344, adding to its direct +0.38), state:17 +0.39 (16902), state:35 +0.30 (13551); state:8/28/3 under 0.05. Evidence: linear replacement AUC 0.8944 (unchanged); mean ablation 0.8935.

Closed form:

$$
\begin{aligned}
y_0 &= \mathrm{relu}\big(-1.83 + 0.984\,\mathrm{relu}\big(1.76 \\
&\qquad + 0.979 x_{state:8}\big)\big) \\
y_1 &= \mathrm{relu}\big(0.806 + 0.987 x_{state:28_b}\big) \\
y_2 &= \mathrm{relu}\big(-0.144 + 1.01 x_{state:3_b}\big) \\
y_3 &= \mathrm{relu}\big(-0.0137 + 0.894 x_{state:47_c}\big) \\
y_4 &= \mathrm{relu}\big(-0.23 + 0.998 x_{state:35}\big) \\
y_5 &= \mathrm{relu}\big(-0.597 + 0.991 x_{state:26_bc}\big) \\
y_6 &= \mathrm{relu}\big(1.04 + 0.997 x_{state:26_bc}\big) \\
y_7 &= \mathrm{relu}\big(0.0457 + 1.0\,\mathrm{relu}\big(1.42 \\
&\qquad + 0.99 x_{state:17} - 0.0231 x_{state:33_b}\big)\big)
\end{aligned}
$$

where y_0 is node `7647`; y_1 is node `8418`; y_2 is node `8896`; y_3 is node `10332`; y_4 is node `13551`; y_5 is node `13725`; y_6 is node `16344`; y_7 is node `16902`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0426, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Units 4967 -> 7647 (state:8), 8418 (state:28), 8896 (state:3), 10332 (state:47), 13551 (state:35), 13725 and 16344 (state:26), 11261 -> 16902 (state:17, state:33). Each depends only on state one-hot indicators; since the states are mutually exclusive, every such function is exactly an offset per state (least-squares R2 = 1.0 for every exit). Hypothesis: a per-state logit offset table; the largest

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.7 A7: service_call_nonlinear

Kind: Composition annotation (depth 1); parent A1 (section 1.6.1); children A8 (section 1.6.8), A9 (section 1.6.9).

- **Entry nodes:** `-71_abde` (numberCustomerServiceCalls_abde), `-59_b` (totalDayMinutes_b), `-60` (totalDayCalls), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-1_b` (state:0_b), `-70_c` (totalIntlCharge_c), `-43_b` (state:47_b), `-71_c` (numberCustomerServiceCalls_c).
- **Exit nodes:** `identity_svc`.
- **Internal nodes:** none.
- **Edges (weight):** `807`→`identity_svc` (1.59), `1264`→`identity_svc` (1.81), `7208`→`identity_svc` (1.52).
- **With its descendants:** 17 nodes.
- **Covers** (coverage, own and as part of the hierarchy): 16 candidate nodes.

Hypothesis. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.59*807 + 1.81*1264 + 1.52*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further (the gate). Evidence: mean-ablating both gives AUC 0.7406, linear replacement of both 0.8313 (all-linear model 0.8314); the gate alone restores 0.8917 of 0.8944; sensitivity of identity_svc on train: day minutes 1.18, state:0 0.61, state:47 0.50, service calls 0.46+0.07, international charge 0.44, evening charge 0.43.

Closed form (children as named functions):

$$
\begin{aligned}
y_0 = {} & 1.81\,\mathrm{csc_usage_gate_1}\big(x_{numberCustomerServiceCalls_abde},\\
& x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b},\\
& x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}\big)\\
& + 1.52\,\mathrm{csc_hinge}\big(x_{numberCustomerServiceCalls_c}\big)\\
& + 1.59\,\mathrm{csc_usage_gate_0}\big(x_{numberCustomerServiceCalls_abde},\\
& x_{totalDayMinutes_b}, x_{totalDayCalls}, x_{totalEveCharge_b}, x_{state:41_b},\\
& x_{state:0_b}, x_{totalIntlCharge_c}, x_{state:47_b}\big)
\end{aligned}
$$

Expanded over the children:

$$
\begin{aligned}
y_0 = {} & 1.81\,\mathrm{relu}\big(0.0266 + 0.329\,\mathrm{relu}\big(-1.3\\
& + 0.584\,\mathrm{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde}\\
& + 0.359\,\mathrm{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big)\\
& + 0.608\,\mathrm{relu}\big(0.951 + 0.362\,\mathrm{relu}\big(-0.199\\
& + 0.917x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls}\\
& - 1.3x_{totalDayMinutes_b}\big) - 0.349x_{totalIntlCharge_c}\\
& - 0.397x_{state:47_b}\big) + 1.52\,\mathrm{relu}\big(-0.695\\
& + 0.559x_{numberCustomerServiceCalls_c}\big) + 1.59\,\mathrm{relu}\big(-1.3\\
& + 0.584\,\mathrm{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde}\\
& + 0.359\,\mathrm{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big)\\
& + 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big)\\
& + 0.608\,\mathrm{relu}\big(0.951 + 0.362\,\mathrm{relu}\big(-0.199
\end{aligned}
$$

$$+ 0.917 x_{numberCustomerServiceCalls_abde})$$
$$+ 0.235 x_{numberCustomerServiceCalls_abde}) + 0.0575 x_{totalDayCalls}$$
$$- 1.3 x_{totalDayMinutes_b})$$

where y_0 is node `identity_svc`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0374, and the exact function by at most 3.55×10^{-15}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Composition of csc_usage_gate and csc_hinge: their exits are summed by identity_svc = 1.59*807 + 1.81*1264 + 1.52*7208, which enters the logit with weight 1; they share only the customer-service-calls input (split copies). Hypothesis: all of the model's service-call effect is nonlinear (calls have no direct edge): risk jumps from 4 calls, and for frequent callers low day usage raises risk further

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.8 A8: csc_usage_gate

Kind: Leaf annotation (depth 2); parent A7 (section 1.6.7).

- **Entry nodes:** `-71_abde` (numberCustomerServiceCalls_abde), `-59_b` (totalDayMinutes_b), `-60` (totalDayCalls), `-64_b` (totalEveCharge_b), `-37_b` (state:41_b), `-1_b` (state:0_b), `-70_c` (totalIntlCharge_c), `-43_b` (state:47_b).
- **Exit nodes:** `807`, `1264`.
- **Internal nodes:** `9555`, `2627`, `15509`, `2796`.
- **Edges (weight):** `-60`→`807` (0.0575), `2627`→`807` (0.608), `2796`→`807` (0.584), `807`→`1264` (0.329), `9555`→`2627` (0.362), `15509`→`2796` (0.359), `-70_c`→`1264` (-0.349), `-64_b`→`2796` (-0.996), `-59_b`→`807` (-1.3), `-43_b`→`1264` (-0.397), `-37_b`→`2796` (0.129), `-1_b`→`2796` (-1.41), `-71_abde`→`2627` (0.235), `-71_abde`→`2796` (0.314), `-71_abde`→`9555` (0.917), `-71_abde`→`15509` (0.99).
- **Covers** (coverage, own and as part of the hierarchy): 12 candidate nodes.

Hypothesis. Service-call x low-day-usage gate. 807 = relu(-1.30 - 1.30 dayMin + 0.06 dayCalls + 0.61 g1(csc) + 0.58 g2), with g1 = relu(0.95 + 0.24 csc + 0.36 relu(0.92 csc - 0.20)) always active (convex in calls) and g2 = 2796 = relu(0.38 + 0.31 csc + 0.36 relu(0.99 csc - 0.20) - 1.00 eveCharge - 1.41 state:0 + 0.13 state:41); 1264 = relu(0.03 + 0.33*807 - 0.35 intlCharge - 0.40 state:47). The logit gets +1.59*807 + 1.81*1264. Hypothesis: 807 is a hinge that switches on at LOW day minutes, and its kink moves up with each customer-service call (median kink on train: 147 day minutes at 0 calls, 157 at 1, 170 at 2, 194 at 3, 215 at 4, 237 at 5, 270 at 6); high evening charge lowers the kink (2796 is on for 98% of customers in the lowest evening-charge band, 22% in the highest) and state:0 switches 2796 off (all 56 customers). Below the kink the gate's slope is about -2.84 logit per SD of day minutes, more than cancelling the direct +1.84, so the logit is V-shaped in day minutes. 1264 passes 807 on and adds a low-international-charge hinge (on for 1319 rows where 807 is off); state:47 switches 1264 off. Evidence (train, n=4000): CSC>=4 and day minutes < 160: 113 customers, empirical churn 0.88, model mean 0.52 (under-predicted); CSC<4 and day minutes < 120: 476 customers, empirical 0.05, model 0.06. Mean churn by calls 0-6: 0.115/0.110/0.110/0.106/0.437/0.571/0.679, model 0.087/0.078/0.080/0.113/0.246/0.503/0.820; share with 807 on rises 0.31 -> 0.93. Sensitivity (807): day minutes 0.57, state:0 0.29, service calls 0.22, evening charge 0.21. Ablation: mean-ablation AUC 0.7690 (full 0.8944); least-squares linear replacement

0.8461 (R2 0.68/0.74); restored alone onto an all-linearised model 0.8917 (all-linear 0.8314). This is the model's main nonlinearity.

Closed form:

$$
\begin{aligned}
y_0 = \operatorname{relu}\big(&-1.3 + 0.584\operatorname{relu}\big(0.382 \\
&+ 0.314x_{numberCustomerServiceCalls_abde} + 0.359\operatorname{relu}\big(-0.2 \\
&+ 0.99x_{numberCustomerServiceCalls_abde}\big) + 0.129x_{state:41_b} \\
&- 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) \\
y_1 = \operatorname{relu}\big(&0.0266 + 0.329\operatorname{relu}\big(-1.3 \\
&+ 0.584\operatorname{relu}\big(0.382 + 0.314x_{numberCustomerServiceCalls_abde} \\
&+ 0.359\operatorname{relu}\big(-0.2 + 0.99x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.129x_{state:41_b} - 1.41x_{state:0_b} - 0.996x_{totalEveCharge_b}\big) \\
&+ 0.608\operatorname{relu}\big(0.951 + 0.362\operatorname{relu}\big(-0.199 \\
&+ 0.917x_{numberCustomerServiceCalls_abde}\big) \\
&+ 0.235x_{numberCustomerServiceCalls_abde}\big) + 0.0575x_{totalDayCalls} \\
&- 1.3x_{totalDayMinutes_b}\big) - 0.349x_{totalIntlCharge_c} \\
&- 0.397x_{state:47_b}\big)
\end{aligned}
$$

where y_0 is node `807`; y_1 is node `1264`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.0112, and the exact function by at most 8.88×10^{-16}.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. Service-call x low-day-usage gate. 807 = relu(-1.30 - 1.30 dayMin + 0.06 dayCalls + 0.61 g1(csc) + 0.58 g2), with g1 = relu(0.95 + 0.24 csc + 0.36 relu(0.92 csc - 0.20)) always active (convex in calls) and g2 = 2796 = relu(0.38 + 0.31 csc + 0.36 relu(0.99 csc - 0.20) - 1.00 eveCharge - 1.41 state:0 + 0.13 state:41); 1264 = relu(0.03 + 0.33*807 - 0.35 intlCharge - 0.40 state:47). The logit gets +1.

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.6.9 A9: csc_hinge

Kind: Leaf annotation (depth 2); parent A7 (section 1.6.7).

- **Entry nodes:** `-71_c` (numberCustomerServiceCalls_c).
- **Exit nodes:** `7208`.
- **Internal nodes:** none.
- **Edges (weight):** `-71_c`→`7208` (0.559).
- **Covers** (coverage, own and as part of the hierarchy): 1 candidate node.

Hypothesis. 7208 = relu(-0.70 + 0.56 csc), logit weight +1.52. Hypothesis: a hinge on customer-service calls that is off for 0-3 calls and switches on from 4 calls (kink at 3.2 calls), adding about +0.65 logit per extra call (contribution +0.52 at 4 calls, +1.17 at 5, +1.82 at 6). Active on exactly the 318 customers with 4+ calls (empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot of 7208 on train rises from 0 to 2.5 (correlation with calls 0.67); mean ablation AUC 0.8901, linear replacement 0.8916; restored alone onto the all-linear model 0.8462 (all-linear 0.8314), but largely redundant once the gate is present (gate alone 0.8917).

Closed form:

$$y_0 = \mathrm{relu}\big(-0.695 + 0.559 x_{numberCustomerServiceCalls_c}\big)$$

where y_0 is node `7208`; z_i, where shown, is the pre-activation of a sigmoid output y_i; weights and biases are shown to 3 significant figures. On the training split (n = 4000), the displayed closed form differs from the forward pass by at most 0.00224, and the exact function by at most 0.

Evidence (2 items; full payloads in the JSON record):

- ANALYTICAL, `analytical_formula`; data split: train. Closed form from get_formula (force).
- EMPIRICAL, `ablation_study`; data split: train. 7208 = relu(-0.70 + 0.56 csc), logit weight +1.52. Hypothesis: a hinge on customer-service calls that is off for 0-3 calls and switches on from 4 calls (kink at 3.2 calls), adding about +0.65 logit per extra call (contribution +0.52 at 4 calls, +1.17 at 5, +1.82 at 6). Active on exactly the 318 customers with 4+ calls (empirical churn jumps from 0.11 at 3 calls to 0.44 at 4). Evidence: line plot o

Authorship: drafted by AI-assisted draft on 2026-11-10; verified by nobody (unverified).

1.7 Audit against the definitions

Automatic findings on the stored explanation (codes as in the JSON record's `issues`); they flag what an examiner should check, not verdicts.

- WARNING `unverified`: No human verification is recorded for this explanation.

end of thesis draft